

Conjecture 00000008565 is false:
Cayley graphs of $\mathbb{Z}/n$, $n \geq 3$, never have simple spectrum

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1 The conjecture

The English text of conjecture 00000008565 reads:

Definition: The spectrum of a Cayley graph of a group G : the integrality of adjacency operator eigenvalues as algebraic integers. Conjecture: All Cayley graph spectra consist of algebraic integers (an algebraic integer law); simple spectrum (no multiplicity) is generic for cyclic generating sets of abelian groups (an abelian simple spectrum law); the lower bound on spectral multiplicities for nonabelian groups is the dimension of irreducible representations (a representation dimension lower bound law); and the extremal spectral norm is given by the growth rate of the ball length function of the group (a ball length extremal law).

The Chinese text states the same four laws; its second clause says that single-multiplicity spectrum (no multiplicity) is generic (universal) for cyclic generating sets of abelian groups.

The conjecture is a conjunction of four laws, so refuting one of them refutes it. We refute the second law, the *abelian simple spectrum law*.

2 Reading and conventions

- **Groups.** We take the cyclic groups $G = \mathbb{Z}/n\mathbb{Z}$ (Lean: `ZMod n`) for every $n \geq 3$. They are abelian, and every generating set of them is a generating set of a cyclic group. The result holds for *every* subset S , so it covers every reading of “cyclic generating set”: any generating set of a cyclic group, a single generator $\{g\}$ or $\{g, -g\}$, and also non-generating sets.
- **Cayley graph.** The Cayley graph is the undirected one. For a symmetric set $S = -S$ with $0 \notin S$, the vertices are the elements of G and $u \sim v$ iff $v - u \in S$. We use Mathlib’s `SimpleGraph.addCayley S`, in which $u \sim v$ iff $u \neq v$ and $(v - u \in S \text{ or } u - v \in S)$. In other words, Mathlib symmetrizes S and drops 0. Lean checks (theorem `addCayley_adj_iff_sub_mem`) that, for symmetric S with $0 \notin S$, this is exactly the textbook relation $v - u \in S$. The spectrum is that of the adjacency matrix $A \in \mathbb{C}^{G \times G}$, $A_{uv} = 1$ if $u \sim v$ and 0 otherwise, acting on $\mathbb{C}^G$. We also treat the textbook matrix $A_{uv} = [v - u \in S]$ for any symmetric S , which allows a loop at every vertex when $0 \in S$ (`connMatrix`).
- **Simple spectrum.** A matrix has simple spectrum if every eigenspace has dimension at most 1 (`HasSimpleSpectrum`). Since A is real symmetric, this is the same as all roots of the characteristic polynomial being simple. We refute both forms: we exhibit an eigenspace of dimension ≥ 2 , and a root of χ_A of multiplicity ≥ 2 .

- **Generic.** Whatever “generic” means (all but finitely many, density or proportion tending to 1, probability 1, open dense), a property that holds for *no* member of a nonempty family is not generic in that family. We prove that for every $n \geq 3$ none of the generating sets of $\mathbb{Z}/n\mathbb{Z}$, which form a nonempty family, gives simple spectrum. The proportion is 0 for every $n \geq 3$.
- **Reading not refuted.** We do not treat *directed* Cayley digraphs with a non-symmetric S and the non-symmetric matrix $A_{uv} = [v - u \in S]$. There the clause may survive. For example, $\text{Cay}(\mathbb{Z}/n, \{1\})$ is a directed n -cycle whose eigenvalues are the n distinct n -th roots of unity. We take the undirected reading because the text speaks of the spectrum, multiplicities and spectral norm of “the adjacency operator”. The retrieved Wikipedia article (Section 6) describes the symmetric case as “a simple undirected graph”.

3 Proof

Fix $n \geq 3$, put $\zeta = e^{2\pi i/n}$, and let $\chi(x) = \zeta^x$ be the standard additive character of $\mathbb{Z}/n$ (Mathlib `ZMod.stdAddChar`). It is injective (`ZMod.injective_stdAddChar`).

Lemma 1 (circulant eigenvectors, `mulVec_of_mul`). *Let $A_{uv} = f(v - u)$ for some $f : \mathbb{Z}/n \rightarrow \mathbb{C}$, and let $\psi : \mathbb{Z}/n \rightarrow \mathbb{C}$ satisfy $\psi(a+b) = \psi(a)\psi(b)$. Then $A\psi = \lambda_f(\psi)\psi$ with $\lambda_f(\psi) = \sum_w f(w)\psi(w)$.*

Proof. $(A\psi)(u) = \sum_v f(v - u)\psi(v) = \sum_w f(w)\psi(u + w) = \psi(u) \sum_w f(w)\psi(w)$, substituting $v = u + w$. $\square$

Lemma 2 (key lemma, `exists_eigenspace_finrank_ge_two_of_kernel`). *If moreover $f(-w) = f(w)$ for all w , then the eigenspace of A for $\mu = \lambda_f(\chi)$ has dimension at least 2.*

Proof. Put $v_1 = \chi$ and $v_2(x) = \chi(-x)$. Both are multiplicative, so by the previous lemma they are eigenvectors with eigenvalues $\lambda_f(v_1)$ and $\lambda_f(v_2)$. Substituting $w \mapsto -w$ and using the evenness of f gives $\lambda_f(v_2) = \sum_w f(w)\chi(-w) = \sum_w f(-w)\chi(w) = \lambda_f(v_1) = \mu$. Suppose $sv_1 + tv_2 = 0$. Evaluating at 0 gives $s + t = 0$. Evaluating at 1 gives $s(\chi(1) - \chi(-1)) = 0$. Since $n \geq 3$ we have $1 \neq -1$ in $\mathbb{Z}/n$ (`ZMod.neg_one_ne_one`), and χ is injective, so $\chi(1) \neq \chi(-1)$. Hence $s = t = 0$. So v_1, v_2 are linearly independent vectors of the μ -eigenspace, and its `finrank` is at least 2. $\square$

Theorem 3 (`exists_eigenspace_finrank_ge_two`, `not_hasSimpleSpectrum`). *For every $n \geq 3$ and every $S \subseteq \mathbb{Z}/n$, the adjacency matrix of $\text{Cay}(\mathbb{Z}/n, S)$ (Mathlib `addCayley`) has an eigenspace of dimension at least 2. Hence it does not have simple spectrum.*

Proof. The Cayley graph is translation invariant: $u \sim u + w$ iff $0 \sim w$ (`addCayley_adj_add_iff_right`). Hence $A_{uv} = A_{0,v-u}$ (`adjMatrix_eq_kernel`). The neighbourhood of 0 is closed under negation: $0 \sim -w$ iff $w \sim 0$ iff $0 \sim w$ (`adj_zero_neg`). Apply the key lemma with $f(w) = A_{0w}$. $\square$

Corollary 4 (`exists_charpoly_rootMultiplicity_ge_two`). *The same μ is a root of the characteristic polynomial of A of multiplicity at least 2. This follows because the geometric multiplicity is at most the algebraic multiplicity (Mathlib `LinearMap.finrank_eigenspace_le` and `Matrix.charpoly_toLin'`).*

Corollary 5 (`connMatrix_not_hasSimpleSpectrum`). *For every $n \geq 3$ and every symmetric $S \subseteq \mathbb{Z}/n$, the textbook matrix $A_{uv} = [v - u \in S]$ does not have simple spectrum. Loops are allowed when $0 \in S$.*

Proof. Apply the key lemma with $f(w) = [w \in S]$, which is even because $S = -S$. $\square$

Corollary 6 (`simple_spectrum_not_generic`). *For every $n \geq 3$, the number of finite sets $S \subseteq \mathbb{Z}/n$ with $\langle S \rangle = \mathbb{Z}/n$ whose Cayley graph has simple spectrum is 0. The number of generating sets is positive, since $\{1\}$ generates (`closure_one_eq_top`).*

For example, $S = \{1, -1\}$ gives the cycle C_n , with eigenvalues $2 \cos(2\pi j/n)$. Each of these with $j \not\equiv -j \pmod{n}$ occurs twice. This matches the 5-cycle example in the retrieved Wikipedia article.

4 Remarks on the other clauses (not formalized)

Clause 1 holds for *finite* Cayley graphs: the eigenvalues of an integer matrix are roots of its monic integer characteristic polynomial. If infinite groups are allowed, as the fourth law’s “growth rate of the ball length function” suggests, clause 1 also fails. For $\text{Cay}(\mathbb{Z}, \{\pm 1\})$ on $\ell^2(\mathbb{Z})$, the Fourier transform turns the adjacency operator into multiplication by $2 \cos \theta$ on L^2 of the circle. Its (full operator) spectrum is therefore $[-2, 2]$, which contains $1/2$, and $1/2$ is not an algebraic integer. Note that this operator has no point spectrum ($1/2$ is not an ℓ^2 eigenvalue), so this backup refutes algebraic integrality of the full spectrum, not a version restricted to eigenvalues. We only sketch this backup and do not formalize it. Clauses 3 and 4 are not addressed.

5 Lean formalization and axiom audit

File `lean/Conjecture8565/Basic.lean` (207 lines, `import Mathlib` only, namespace `C8565`). It was built with `lake build` (Lean toolchain and `Mathlib` as in `lean-toolchain` and `lake-manifest.json`). The main statements are:

- `not_hasSimpleSpectrum (hn : 3 <= n) (S : Set (ZMod n)) [DecidableRel (addCayley S).Adj] : not HasSimpleSpectrum ((addCayley S).adjMatrix C);`
- `exists_eigenspace_finrank_ge_two: there is  $\mu$  with  $2 \leq \text{finrank } C$  (eigenspace (toLin' A)  $\mu$ );`
- `exists_charpoly_rootMultiplicity_ge_two: there is  $\mu$  with  $2 \leq A.\text{charpoly}.\text{rootMultiplicity } \mu$ ;`
- `connMatrix_not_hasSimpleSpectrum`, for the textbook matrix with symmetric S ;
- `simple_spectrum_not_generic`, the counting statement over generating Finsets;
- `addCayley_adj_iff_sub_mem`, the faithfulness check of the Cayley graph.

Here `HasSimpleSpectrum A` is `forall mu, finrank C (eigenspace (toLin' A) mu) <= 1`. `#print axioms` reports `[propext, Classical.choice, Quot.sound]` for each of the first five, and `[propext, Quot.sound]` for the last. There is no `sorry`, no `native_decide` and no added axiom.

6 Sources

- Wikipedia, “Cayley graph”, https://en.wikipedia.org/wiki/Cayley_graph (retrieved 2026-10-05). Verbatim: “In this case, the uncolored Cayley graph can be represented as a simple undirected graph.” The case meant is a symmetric S without the identity. Also: “Every group character ... induces an eigenvector of the adjacency matrix”, and “It is interesting to note that this eigenbasis is independent of the generating set”.
- Mathlib: `SimpleGraph.addCayley` (Mathlib/Combinatorics/SimpleGraph/Cayley.lean, by Edward van de Meent); `ZMod.stdAddChar` (Mathlib/Analysis/SpecialFunctions/Complex/CircleAddChar.lean); `LinearMap.finrank_eigenspace_1e` (Mathlib/LinearAlgebra/Eigenspace/Zero.lean).